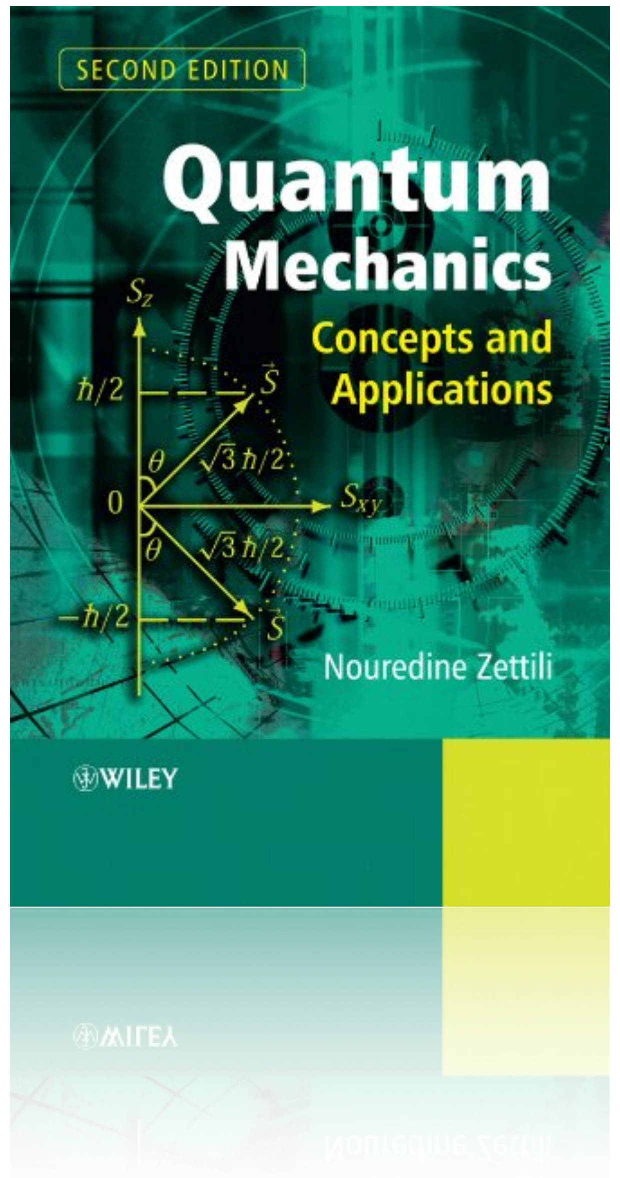
Some useful mathematical concepts

- Hilbert space
- Dirac notation
- Operators
- Eigenvalues and eigenvectors



Hilbert space

A generalisation of the well-known Euclidian space $\mathbb{R}^3$ to spaces with any finite or infinite number of dimensions

A Hilbert space is a a linear vector space containing vectors $\{\psi_a, \psi_b, \psi_c, \ldots\}$ and scalars $\{a, b, c, \ldots\}$ characterised by:

1. rules for vector addition and scalar multiplication

- if ϕ and ψ are elements of $\mathcal{H}$, than $\phi + \psi$ is also an element of $\mathcal{H}$
- $\phi + \psi = \psi + \phi$ (communitivity)
- $(\phi + \psi) + \chi = \phi + (\psi + \chi)$ (associativity)
- existence of zero (or neutral) element: $\phi + 0 = 0 + \phi = \phi$
- existence of an inverse element: $\phi + (-\phi) = 0$
- if ϕ is an element of $\mathcal{H}$, then $a\phi$ also belongs to $\mathcal{H}$ (a is a scalar)
- $(a + b)\psi = a\psi + b\psi$ and $a(\phi + \psi) = a\phi + a\psi$ (distributivity)
- $a(b\psi) = (ab)\psi$ (associativity)
- $I\psi = \psi I = \psi$ and $0\psi = \psi 0 = 0$ (unitary and zero element)

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1. rules for vector addition and scalar multiplication
2. the existence of a scalar product of any of the elements of $\mathcal{H}$, $(\phi, \psi) \in \mathbb{C}$
   - hermiticity $(\phi, \psi) = (\psi, \phi)^*$
   - linearity $(\phi, a\psi_1 + b\psi_2) = a(\phi, \psi_1) + b(\phi, \psi_2)$
   - antilinearity $(c_1\phi_1 + c_2\phi_2, \psi) = c_1^*(\phi_1, \psi) + c_2^*(\phi_2, \psi)$
   - $(\phi, \phi) \geq 0$ (the equality holds for $\phi = 0$)

The existence of a scalar product implies a norm: $||\phi|| = \sqrt{(\phi, \phi)}$

and provides a measure for the distance between two vectors: $d(\phi, \psi) = ||\phi - \psi||$

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3. being separable. That is, the space has an orthonormal basis
4. **being complete. Each element of the Hilbert can be represented by a series of (in)finite length of elements of the Hilbert space.**

Dirac notation

$\{\phi, \upsilon, \psi, \chi\}$ ➡ elements of a Hilbert space: also known as <u>state vectors</u>

$\phi(x,y,z) = \phi(r,\phi,\psi) = \phi(\rho,\phi,z) = \boxed{|\phi\rangle \longrightarrow \text{ket}}$ state vector

$\boxed{\langle\psi| \longrightarrow \text{bra}}$ transposed complex conjugate of $|\phi\rangle$

$\boxed{\langle\psi|\phi\rangle \longrightarrow \text{bra-ket}}$ scalar (or inner) product of the two state vectors

Some useful properties:

$$|\psi\rangle \quad \leftrightarrow \quad \langle\psi|$$

$$a|\psi\rangle + b|\phi\rangle \quad \leftrightarrow \quad a^*\langle\psi| + b^*\langle\phi|$$

$$|a\psi\rangle = a|\psi\rangle, \quad \langle a\psi| = a^*\langle\psi|$$

$$\langle\phi|\psi\rangle^* = \langle\psi|\phi\rangle$$

$$\langle\phi|a\psi + b\chi\rangle = a\langle\phi|\psi\rangle + b\langle\phi|\chi\rangle$$

$$\langle a\psi + b\chi|\phi\rangle = a^*\langle\psi|\phi\rangle + b^*\langle\chi|\phi\rangle$$

$$\langle a_1\psi_1 + b_1\phi_1|a_2\psi_2 + b_2\phi_2\rangle = a_1^*a_2\langle\psi_1|\psi_2\rangle + a_1^*b_2\langle\psi_1|\phi_2\rangle + b_1^*a_2\langle\phi_1|\psi_2\rangle + b_1^*b_2\langle\phi_1|\phi_2\rangle$$

$\langle\phi|\phi\rangle =$ real, positive number

Dirac notation

Schwartz inequality: $|\langle\phi|\chi\rangle|^2 \leq \langle\phi|\phi\rangle\langle\chi|\chi\rangle$ $\langle\mathbf{v},\mathbf{u}\rangle^2 \leq \|\mathbf{v}\|\cdot\|\mathbf{u}\|$

Triangle inequality: $\sqrt{\langle\phi+\chi|\phi+\chi\rangle} \leq \sqrt{\langle\phi|\phi\rangle} + \sqrt{\langle\chi|\chi\rangle}$ $|\mathbf{v}+\mathbf{u}| \leq |\mathbf{v}| + |\mathbf{u}|$

Orthogonal states: $\langle\psi|\phi\rangle = 0$

Orthonormal states: $\langle\psi|\phi\rangle = 0$ and $\langle\phi|\phi\rangle = 1$ $\langle\psi|\psi\rangle = 1$

Forbidden: $\langle\psi|\langle\phi|$ and $|\psi\rangle|\phi\rangle$ when ϕ and ψ belong to the same vector space

otherwise $|\psi\rangle \otimes |\phi\rangle$ defines a new vector space

$\psi(x, y, z)$ $\phi(\sigma)$

cartesian coordinates spin coordinates

Schwartz inequality

For vectors: $\langle \mathbf{v}, \mathbf{u} \rangle^2 \leq \|\mathbf{v}\| \cdot \|\mathbf{u}\|$

For complex functions: $\boxed{|\langle \phi | \chi \rangle|^2 \leq \langle \phi | \phi \rangle \langle \chi | \chi \rangle}$

$$\psi = \phi + \lambda \chi$$

$$\langle \psi | \psi \rangle = \langle \phi | \phi \rangle + \lambda \langle \phi | \chi \rangle + \lambda^* \langle \chi | \phi \rangle + \lambda \lambda^* \langle \chi | \chi \rangle \geq 0$$

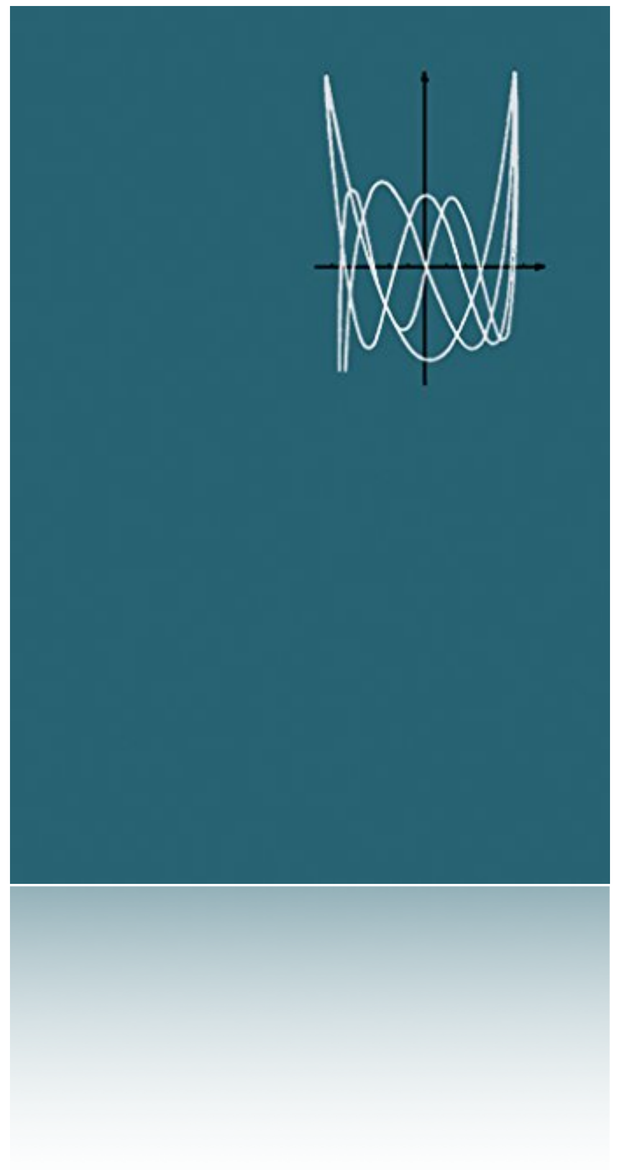
$$\frac{\partial}{\partial \lambda^*} \langle \psi | \psi \rangle = \langle \chi | \phi \rangle + \lambda \langle \chi | \chi \rangle = 0 \qquad \frac{\partial}{\partial \lambda} \langle \psi | \psi \rangle = \langle \phi | \chi \rangle + \lambda^* \langle \chi | \chi \rangle = 0$$

$$\lambda = -\frac{\langle \chi | \phi \rangle}{\langle \chi | \chi \rangle} \qquad \lambda * = -\frac{\langle \phi | \chi \rangle}{\langle \chi | \chi \rangle}$$

$$\langle \psi | \psi \rangle = \langle \phi | \phi \rangle - \frac{\langle \chi | \phi \rangle \langle \phi | \chi \rangle}{\langle \chi | \chi \rangle} - \frac{\langle \phi | \chi \rangle \langle \chi | \phi \rangle}{\langle \chi | \chi \rangle} + \frac{\langle \chi | \phi \rangle \langle \phi | \chi \rangle}{\langle \chi | \chi \rangle^2} \langle \chi | \chi \rangle \geq 0$$

$$\langle \phi | \phi \rangle \langle \chi | \chi \rangle \geq |\langle \phi | \chi \rangle|^2$$

Arfken
Mathematical methods for Physicists



Operators

Operator $\hat{A}$ is a mathematical rule that converts $|\psi\rangle$ into $|\psi'\rangle$ ➡

- differentiation
- multiplication
- permutation

$$\hat{A}|\psi\rangle = |\psi'\rangle$$

same for bra-vectors

$$\langle\phi|\hat{A} = \langle\phi'|$$

unity operator: $\hat{I}|\psi\rangle = |\psi\rangle$

gradient: $\nabla\psi(x,y,z) = \frac{\partial}{\partial x}\psi(x,y,z) + \frac{\partial}{\partial y}\psi(x,y,z) + \frac{\partial}{\partial z}\psi(x,y,z)$

linear momentum: $\hat{P} = -i\hbar\nabla$

Laplacian operator: $\nabla^2\psi(x,y,z) = \frac{\partial^2}{\partial x^2}\psi(x,y,z) + \frac{\partial^2}{\partial y^2}\psi(x,y,z) + \frac{\partial^2}{\partial z^2}\psi(x,y,z)$

linear operator: $\hat{A} : \mathcal{H} \to \mathcal{H}$

$\hat{A}\psi = \phi \in \mathcal{H} \quad \forall \quad \psi \in \mathcal{H}$

implying $\hat{A}(\psi + \phi) = \hat{A}\psi + \hat{A}\phi$ and $\hat{A}\lambda\psi = \lambda\hat{A}\psi$

Operators

order is important: $\hat{A}\hat{B} \neq \hat{B}\hat{A}$

if $\hat{B}|\psi\rangle = |\psi'\rangle$ then $\hat{A}\hat{B}|\psi\rangle = \hat{A}\left(\hat{B}|\psi\rangle\right) = \hat{A}|\psi'\rangle$

expectation value: $\langle\hat{A}\rangle = \dfrac{\langle\psi|\hat{A}|\psi\rangle}{\langle\psi|\psi\rangle}$

$|\psi\rangle\langle\phi|$ is an operator: $|\psi\rangle\langle\phi|\ |\chi\rangle = \langle\phi|\chi\rangle|\psi\rangle = |\psi'\rangle$

Hermitian operator: $\hat{A}^\dagger = \hat{A}$ or also, $\langle\psi|\hat{A}^\dagger|\phi\rangle = \langle\phi|\hat{A}|\psi\rangle^*$

expectation value: $\langle\phi|\hat{A}^\dagger|\phi\rangle = \langle\phi|\hat{A}|\phi\rangle^*$ ➡ real number

anti-Hermitian operator: $\hat{A}^\dagger = -\hat{A}$ or also, $\langle\psi|\hat{A}^\dagger|\phi\rangle = -\langle\phi|\hat{A}|\psi\rangle^*$

expectation value: $\langle\phi|\hat{A}^\dagger|\phi\rangle = -\langle\phi|\hat{A}|\phi\rangle^*$ ➡ imaginary number

Commutators

$[\hat{A},\hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ commutator

if $[\hat{A},\hat{B}] = 0$ then $\hat{A}$ and $\hat{B}$ commute

$\{\hat{A},\hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$ anti-commutator

$[\hat{A},\hat{A}] = 0$

For example: $[\hat{x},\hat{P}_x] = i\hbar$

$$\hat{x}\hat{P}_x\psi(r) = -i\hbar x\frac{\partial\psi(r)}{\partial x}$$

$$\hat{P}_x\hat{x}\psi(r) = -i\hbar\frac{\partial}{\partial x}x\psi(r) = -i\hbar\psi(r) - i\hbar x\frac{\partial\psi(r)}{\partial x}$$

$\Rightarrow$ $[\hat{x},\hat{P}_x]\psi(r) = i\hbar\psi(r)$

Anti-symmetric: $[\hat{A},\hat{B}] = -[\hat{B},\hat{A}]$

Distributive: $[\hat{A},\hat{B}\hat{C}] = [\hat{A},\hat{B}]\hat{C} + \hat{B}[\hat{A},\hat{C}]$

$[\hat{A}\hat{B},\hat{C}] = \hat{A}[\hat{B},\hat{C}] + [\hat{A},\hat{C}]\hat{B}$

Operators commute with numbers: $[\hat{A},b] = 0$

Heisenberg's uncertainty relations

$$\Delta\hat{A} = \hat{A} - \langle\hat{A}\rangle$$

$$\Delta\hat{B} = \hat{B} - \langle\hat{B}\rangle$$

$$(\Delta\hat{A})^2 = \hat{A}^2 - 2\hat{A}\langle\hat{A}\rangle + \langle\hat{A}\rangle^2$$

$$(\Delta\hat{B})^2 = \hat{B}^2 - 2\hat{B}\langle\hat{B}\rangle + \langle\hat{B}\rangle^2$$

$$\langle\psi|(\Delta\hat{A})^2|\psi\rangle = \langle\psi|\hat{A}^2|\psi\rangle - 2\langle\psi|\hat{A}|\psi\rangle\langle\hat{A}\rangle + \langle\hat{A}\rangle^2$$

$$\langle(\Delta\hat{A})^2\rangle = \langle\hat{A}^2\rangle - 2\langle\hat{A}\rangle\langle\hat{A}\rangle + \langle\hat{A}\rangle^2$$

$$\langle(\Delta\hat{A})^2\rangle = \langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2$$

$$\langle(\Delta\hat{B})^2\rangle = \langle\hat{B}^2\rangle - \langle\hat{B}\rangle^2$$

$$\boxed{\Delta A = \sqrt{\langle(\Delta\hat{A})^2\rangle}} = \sqrt{\langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2}$$

$$\boxed{\Delta B = \sqrt{\langle(\Delta\hat{B})^2\rangle}} = \sqrt{\langle\hat{B}^2\rangle - \langle\hat{B}\rangle^2}$$

Heisenberg's uncertainty relations

$\chi = \Delta\hat{A}\psi = \left(\hat{A} - \langle\hat{A}\rangle\right)\psi$

$\phi = \Delta\hat{B}\psi = \left(\hat{B} - \langle\hat{B}\rangle\right)\psi$

$\hat{A}$ and $\hat{B}$ are Hermitian, therefore $\Delta\hat{A}$ and $\Delta\hat{B}$ are also Hermitian

$\langle\chi|\chi\rangle = \langle\psi|\Delta\hat{A}^\dagger\Delta\hat{A}|\psi\rangle = \langle\psi|(\Delta\hat{A})^2|\psi\rangle = \langle(\Delta\hat{A})^2\rangle$

$\langle\phi|\phi\rangle = \langle(\Delta\hat{B})^2\rangle$

$\langle\phi|\chi\rangle = \langle\Delta\hat{A}\Delta\hat{B}\rangle$

Heisenberg's uncertainty relations

$$\chi = \Delta\hat{A}\psi = \left(\hat{A} - \langle\hat{A}\rangle\right)\psi \qquad\qquad \phi = \Delta\hat{B}\psi = \left(\hat{B} - \langle\hat{B}\rangle\right)\psi$$

$\hat{A}$ and $\hat{B}$ are Hermitian, therefore $\Delta\hat{A}$ and $\Delta\hat{B}$ are also Hermitian

$$\langle\chi|\chi\rangle = \langle\psi|\Delta\hat{A}^\dagger\Delta\hat{A}|\psi\rangle = \langle\psi|(\Delta\hat{A})^2|\psi\rangle = \langle(\Delta\hat{A})^2\rangle$$

$$\langle\phi|\phi\rangle = \langle(\Delta\hat{B})^2\rangle$$

$$\langle\phi|\chi\rangle = \langle\Delta\hat{A}\Delta\hat{B}\rangle$$

$$\langle\phi|\phi\rangle\langle\chi|\chi\rangle \geq |\langle\phi|\chi\rangle|^2$$

$$\Rightarrow \quad \langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \geq |\langle\Delta\hat{A}\Delta\hat{B}\rangle|^2$$

Heisenberg's uncertainty relations

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \geq |\langle\Delta\hat{A}\Delta\hat{B}\rangle|^2$$

$$\Delta\hat{A}\Delta\hat{B} = \frac{1}{2}[\Delta\hat{A},\Delta\hat{B}] + \frac{1}{2}\{\Delta\hat{A},\Delta\hat{B}\}$$

$$[\Delta\hat{A},\Delta\hat{B}] = \Delta\hat{A}\Delta\hat{B} - \Delta\hat{B}\Delta\hat{A}$$

$$= \hat{A}\hat{B} - \hat{A}\langle\hat{B}\rangle - \langle\hat{A}\rangle\hat{B} + \langle\hat{A}\rangle\langle\hat{B}\rangle$$

$$-\hat{B}\hat{A} + \hat{B}\langle\hat{A}\rangle + \langle\hat{B}\rangle\hat{A} - \langle\hat{B}\rangle\langle\hat{A}\rangle$$

$$[\Delta\hat{A},\Delta\hat{B}] = [\hat{A},\hat{B}]$$

$$\left|\langle\Delta\hat{A}\Delta\hat{B}\rangle\right|^2 = \frac{1}{4}\left|\langle[\hat{A},\hat{B}]\rangle\right|^2 + \frac{1}{4}\left|\langle\{\Delta\hat{A},\Delta\hat{B}\}\rangle\right|^2$$

$$\left|\langle\Delta\hat{A}\Delta\hat{B}\rangle\right|^2 \geq \frac{1}{4}\left|\langle[\hat{A},\hat{B}]\rangle\right|^2$$

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \geq \left|\langle\Delta\hat{A}\Delta\hat{B}\rangle\right|^2 \geq \frac{1}{4}\left|\langle[\hat{A},\hat{B}]\rangle\right|^2$$

Heisenberg's uncertainty relations

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \geq \left|\langle\Delta\hat{A}\Delta\hat{B}\rangle\right|^2 \geq \frac{1}{4}\left|\langle[\hat{A},\hat{B}]\rangle\right|^2$$

$$\Delta A = \sqrt{\langle(\Delta\hat{A})^2\rangle} \qquad \Delta B = \sqrt{\langle(\Delta\hat{B})^2\rangle}$$

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \geq \frac{1}{4}\left|\langle[\hat{A},\hat{B}]\rangle\right|^2$$

$$\Downarrow$$

$$\boxed{\Delta A\Delta B \geq \frac{1}{2}|\langle[\hat{A},\hat{B}]\rangle|}$$

$$\Delta x\Delta p_x \geq \frac{1}{2}|\langle[\hat{x},\hat{p_x}]\rangle|$$

$$\Downarrow \qquad [\hat{x},\hat{p_x}] = i\hbar$$

$$\Delta x\Delta p_x \geq \frac{\hbar}{2}$$

$$\Delta y\Delta p_y \geq \frac{\hbar}{2}$$

$$\Delta z\Delta p_z \geq \frac{\hbar}{2}$$

Eigenvalues and Eigenvectors

$\hat{A}|\psi\rangle = a|\psi\rangle \longrightarrow$ eigenvalue problem

$\hat{A} \longrightarrow$ operator

$|\psi\rangle \longrightarrow$ state vector = eigenvector

$a \longrightarrow$ complex number = eigenvalue

Hermitian operator: $\boxed{\text{eigenvalues are real, eigenvectors are orthogonal}}$

$\hat{A} = \hat{A}^\dagger \;\; ; \;\; \hat{A}|\psi_n\rangle = a_n|\psi_n\rangle \qquad \Rightarrow a_n$ is a real number, and $\langle\psi_n|\psi_m\rangle = 0$

$$\hat{A}|\psi_n\rangle = a_n|\psi_n\rangle \quad \Rightarrow \quad \langle\psi_m|\hat{A}|\psi_n\rangle = a_n\langle\psi_m|\psi_n\rangle$$

$$\langle\psi_m|\hat{A}^\dagger = a_m^*\langle\psi_m| \quad \Rightarrow \quad \langle\psi_m|\hat{A}^\dagger|\psi_n\rangle = a_m^*\langle\psi_m|\psi_n\rangle$$

$$\Rightarrow \quad 0 = (a_n - a_m^*)\langle\psi_m|\psi_n\rangle$$

$m = n$	$\langle\psi_n|\psi_n\rangle > 0$	$a_n = a_n^*$
$m \neq n$	$a_n \neq a_m^*$	$\langle\psi_m|\psi_n\rangle = 0$

Eigenvalues and Eigenvectors

Hermitian operator: eigenvalues are real, eigenvectors are orthogonal

$\boxed{\text{eigenvectors form an orthogonal complete basis}}$

$$\phi = \sum_n c_n \psi_n$$ with ψ_n eigenvectors of $\hat{A}$

orthonormality: $\langle \phi_m | \phi_n \rangle = \delta_{nm}$

completeness: $\sum_{i=1}^{\infty} |\psi_i\rangle\langle\psi_i| = \hat{I}$

Eigenvalues and Eigenvectors

Hermitian operator: eigenvalues are real, eigenvectors are orthogonal

eigenvectors form an orthogonal complete basis

if $[\hat{A}, \hat{B}] = 0$, then the eigenvectors of $\hat{A}$ are also eigenvectors of $\hat{B}$

$$\hat{A}|\phi_n\rangle = a_n|\phi_n\rangle$$

$$\hat{A}|\phi_m\rangle = a_m|\phi_m\rangle$$

$$a_n \neq a_m$$

$$\langle\phi_m|[\hat{A}, \hat{B}]|\phi_n\rangle = \langle\phi_m|\hat{A}\hat{B}|\phi_n\rangle - \langle\phi_m|\hat{B}\hat{A}|\phi_n\rangle = 0$$

(commutator is zero)

$$a_m\langle\phi_m|\hat{B}|\phi_n\rangle - a_n\langle\phi_m|\hat{B}|\phi_n\rangle = 0$$

$$(a_m - a_n)\langle\phi_m|\hat{B}|\phi_n\rangle = 0$$

$$\langle\phi_n|\hat{B}|\phi_m\rangle \propto \delta_{nm} \quad \Rightarrow \quad \hat{B}|\phi_n\rangle = b_n|\phi_n\rangle$$

Eigenvalues and Eigenvectors

Hermitian operator: eigenvalues are real, eigenvectors are orthogonal

eigenvectors form an orthogonal complete basis

if $[\hat{A}, \hat{B}] = 0$, then the eigenvectors of $\hat{A}$ are also eigenvectors of $\hat{B}$

any linear combination of degenerate eigenvectors of $\hat{A}$ is also an eigenvector of $\hat{A}$

$$\hat{A}|\psi_n\rangle = a_n|\psi_n\rangle \qquad \hat{A}|\psi_m\rangle = a_m|\psi_m\rangle \qquad a_m = a_n$$

$$\hat{A}|c_1\psi_n + c_2\psi_m\rangle = c_1\hat{A}|\psi_n\rangle + c_2\hat{A}|\psi_m\rangle = c_1 a_n|\psi_n\rangle + c_2 a_m|\psi_m\rangle$$

$$= a_n\big(c1|\psi_n\rangle + c_2|\psi_m\rangle\big)$$